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WORD LINE PROTECTION METHOD IN THE BACKSIDE PROCESS OF A VERTICAL DYNAMIC RANDOM ACCESS MEMORY (DRAM) DEVICE

Abstract

A semiconductor memory device includes trench isolations arranged in a bit-line direction, gate structures arranged in a word-line direction perpendicular to the bit-line direction, an array of vertical-transistor channels arranged in a vertical direction perpendicular to the bit-line direction and the word-line direction and separated by the trench isolations and gate structures, top ends of the array of the vertical-transistor channels in each column being connected to a line of semiconductor structure extending in the bit-line direction at a backside of the semiconductor memory device, and air gap tunnels along the word-line direction that each crosses below the line of semiconductor structure and between two neighboring vertical-transistor channels in a first region at the backside of the semiconductor memory device.

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Background/Summary

INCORPORATION BY REFERENCE
[0001] The present application is a bypass continuation application of International Application No. PCT/CN2024/079207, filed on Feb. 29, 2024, which is incorporated by reference herein in its entirety.
TECHNICAL FIELD
[0002] The present disclosure relates to a semiconductor device and the fabrication process thereof. The semiconductor device can be a vertical gate dynamic random access memory (DRAM) device.
BACKGROUND
[0003] Planar memory cells are scaled to smaller sizes by improving process technology, circuit design, programming algorithm, and fabrication process. However, as feature sizes of the memory cells approach a lower limit, planar process and fabrication techniques become challenging and costly. As a result, memory density for planar memory cells approaches an upper limit. A three-dimensional (3D) memory architecture can be employed to address the density limitation in planar memory cells by using vertical gate transistors.
SUMMARY
[0004] Aspects of the disclosure provide a method of fabricating a semiconductor memory device. The method can include thinning the semiconductor memory device from a backside of the semiconductor device that, at current stage, has trench isolations formed in a bit-line direction, gate structures formed in a word-line direction perpendicular to the bit-line direction, and an array of vertical-transistor channel structures that extend in a vertical direction perpendicular to the bit-line direction and the word-line direction and are separated by the trench isolations and the gate structures, wherein after the thinning, the trench isolations are exposed at the back of the semiconductor device, top ends of the vertical-transistor channel structures in each column along the bit-line direction are connected with a line of semiconductor structure extending along the bit-line direction at the backside of the semiconductor memory device to form a bridge-shaped structure, and the trench isolations separate neighboring bridge-shaped structures, recessing, in a first region of the backside of the semiconductor memory device, the trench isolations between the neighboring bridge-shaped structures to expose the gate structures formed in the word-line direction, etching the gate structures formed in the word-line direction, resulting in tunnels along the respective gate structures, the tunnels crossing below each line of semiconductor structure, filling a space where the trench isolations and the gate structures are etched away with a sacrificial material, covering a portion of the first region of the backside of the semiconductor memory device with an insulation layer, the insulation layer enclosing the sacrificial materials below the insulation layer and between the respective neighboring bridge-shaped structures at the covered portion of the first region, and removing the sacrificial material within the space where the trench isolations and the gate structures are etched away while the portion of the first region is covered with the

insulation layer, wherein the sacrificial material enclosed by the insulation layer is removed through the tunnels crossing below respective lines of semiconductor structures.

[0005] In one embodiment, the insulation layer covers a second region neighboring the portion of the first region covered by the insulation layer, and the trench isolations in the second region are maintained while the trench isolations in the first region are etched.

[0006] In one embodiment, the semiconductor memory device has metal shields formed between every two neighboring channel structures along the word-line direction.

[0007] In one embodiment, the method further includes etching the metal shields formed in the word-line direction, resulting in tunnels along the respective metal shield crossing below each line of semiconductor structures, filling the space where the trench isolations and the metal shields are etched away with the sacrificial material, and removing the sacrificial material within the space where the trench isolations and the metal shields are etched away while the portion of the first region is covered with the insulation layer, wherein the sacrificial material enclosed by the insulation layer is removed through the tunnels along the respective metal shields crossing below respective lines of semiconductor structures.

[0008] In one embodiment, the method further includes covering surfaces of the space where the trench isolations and the gate structures are etched away with a spacer layer before filling the space where the trench isolations and the gate structures are etched away with the sacrificial material.

[0009] In one embodiment, the method further includes forming bit-line structures over the lines of semiconductor structures not covered by the insulation layer in the first region while the portion of the first region is covered with the insulation layer.

[0010] Aspects of the disclosure provide a semiconductor memory device. The semiconductor memory device includes trench isolations arranged in a bit-line direction, gate structures arranged in a word-line direction perpendicular to the bit-line direction, an array of vertical-transistor channels arranged in a vertical direction perpendicular to the bit-line direction and the word-line direction and separated by the trench isolations and gate structures, top ends of the array of the vertical-transistor channels in each column being connected to a line of semiconductor structure extending in the bit-line direction at a backside of the semiconductor memory device, and air gap tunnels along the word-line direction that each crosses below the line of semiconductor structure and between two neighboring vertical-transistor channels in a first region at the backside of the semiconductor memory device.

[0011] Aspects of the disclosure provide a memory system includes a memory controller, and a semiconductor memory device coupled to the memory controller. The semiconductor memory device includes trench isolations arranged in a bit-line direction, gate structures arranged in a word-line direction perpendicular to the bit-line direction, an array of vertical-transistor channels arranged in a vertical direction perpendicular to the bit-line direction and the word-line direction and separated by the trench isolations and gate structures, top ends of the array of the vertical-transistor channels in each column being connected to a line of semiconductor structure extending in the bit-line direction at a backside of the semiconductor memory device, and air gap tunnels along the word-line direction that each crosses below the line of semiconductor structure and between two neighboring vertical-transistor channels in a first region at the backside of the semiconductor memory device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Aspects of the present disclosure can be understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be increased or reduced for clarity of discussion.

[0013] FIG. **1** illustrates a block diagram of an exemplary system having a memory device, according to some aspects of the present disclosure.

[0014] FIG. **2** illustrates a schematic diagram of a memory device including peripheral circuits and an array of memory cells each having a vertical transistor, according to some aspects of the present disclosure.

[0015] FIG. **3** illustrates a schematic circuit diagram of a memory device including peripheral circuits and an array of dynamic random-access memory (DRAM) cells, according to some aspects of the present disclosure.

[0016] FIG. **4** illustrates a schematic circuit diagram of a memory device including peripheral circuits and an array of phase-change memory (PCM) cells, according to some aspects of the present disclosure.

[0017] FIGS. **5**A-**5**B illustrate a fabrication result of overlapping of trench recess regions and insulating layers for bit line formation at a backside of the semiconductor device.

[0018] FIGS. **6**A-**6**B illustrate a fabrication result of non-overlapping of trench recess regions and insulating layers for bit line formation at a backside of the semiconductor device.

[0019] FIGS. **7**A-**1**˜**7**A-**2**/7B-**1**˜**7**B-**2**/7C-**1**˜**7**C-**2**/7D-**1**˜**7**D-**5**/7E/**7**F-**1**˜**7**F-**5**/7G-**1**˜**7**G-**7**/7H-**1**˜**7**H-**7**/7I-**1**˜**7**I-**7**/7J-**1**˜**7**J-**7**/7K-**1**˜**7**K-**7** illustrate a fabrication process for forming a memory device according to some aspects of the present disclosure.

[0020] FIG. **8** illustrates a flowchart of a fabrication process **800** of forming a semiconductor memory device according to embodiments of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

[0021] Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. As such, other configurations and arrangements can be used without departing from the scope of the present disclosure. Also, the present disclosure can also be employed in a variety of other applications. Functional and structural features as described in the present disclosures can be combined, adjusted, and modified with one another and in ways not specifically depicted in the drawings, such that these combinations, adjustments, and modifications are within the scope of the present disclosure.

[0022] In general, terminology may be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, may be used to describe any feature, structure, or characteristic in a singular sense or may be used to describe combinations of features, structures or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, may be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” may be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for the existence of additional factors not necessarily expressly described, again, depending at least in part on context.

[0023] It should be readily understood that the meaning of “on,” “above,” and “over” in the present disclosure should be interpreted in the broadest manner such that “on” not only means “directly on” something but also includes the meaning of “on” something with an intermediate feature or a layer therebetween, and that “above” or “over” not only means the meaning of “above” or “over” something but can also include the meaning it is “above” or “over” something with no intermediate feature or layer therebetween (directly on something).

[0024] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0025] As used herein, the term “substrate” refers to a material onto which subsequent material layers are added. The substrate itself can be patterned. Materials added on top of the substrate can be patterned or can remain unpatterned. Furthermore, the substrate can include a wide array of semiconductor materials, such as silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate can be made from an electrically non-conductive material, such as a glass, a plastic, or a sapphire wafer.

[0026] As used herein, the term “layer” refers to a material portion including a region with a thickness. A layer can extend over the entirety of an underlying or overlying structure or may have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any pair of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layers thereupon, thereabove, and/or therebelow. A layer can include multiple layers. For example, an interconnect layer can include one or more conductors and contact layers (in which interconnect lines and/or vertical interconnect access (via) contacts are formed) and one or more dielectric layers.

I. Memory Devices with Vertical Transistors

[0027] Transistors are used as the switch or selecting devices in the memory cells of some memory devices, such as dynamic random access memory (DRAM), phase-change memory (PCM), and ferroelectric DRAM (FRAM). However, the planar transistors commonly used in existing memory cells usually have a horizontal structure with buried word lines in the substrate and bit lines above the substrate. Since the source and drain of a planar transistor are disposed laterally at different locations, which increases the area occupied by the transistor. The design of planar transistors also complicates the arrangement of interconnected structures, such as word lines and bit lines, coupled to the memory cells, for example, limiting the pitches of the word lines and/or bit lines, thereby increasing the fabrication complexity and reducing the production yield. Moreover, because the bit lines and the storage units (e.g., capacitors or PCM elements) are arranged on the same side of the planar transistors (above the transistors and substrate), the bit line process margin is limited by the storage units, and the coupling capacitance between the bit lines and storage units, such as capacitors, are increased. Planar transistors may also suffer from a high leakage current as the saturated drain current keeps increasing, which is undesirable for the performance of memory devices.

[0028] On the other hand, the memory cell array and the peripheral circuits for controlling the memory cell array are usually arranged side-by-side in the same plane. As the number of memory cells keeps increasing, to maintain the same chip size, the dimensions of the components in the memory cell array, such as transistors, word lines, and/or bit lines, need to keep decreasing in order not to significantly reduce the memory cell array efficiency.

[0029] To address one or more of the aforementioned issues, vertical transistors can replace the planar transistors as the switch and selecting devices in a memory cell array of memory devices (e.g., DRAM, PCM, and FRAM). Compared with planar transistors, the vertically arranged transistors (e.g., the drain and source are overlapped in the plan view) can reduce the area of the transistor as well as simplify the layout of the interconnect structures, e.g., metal wiring the word lines and bit lines, which can reduce the fabrication complexity and improve the yield. For example, the pitches of word lines and/or bit lines can be reduced for case of fabrication. The vertical structures of the transistors also allow the bit lines and storage units, such as capacitors, to be arranged on opposite sides of the transistors in the vertical direction (e.g., one above and on below the transistors), such that the process margin of the bit lines can be increased and the coupling capacitance between the bit lines and the storage units can be decreased.

[0030] Also, the memory cell array having vertical transistors and the peripheral circuits of the memory cell array can be formed on different wafers and bonded together in a face-to-face manner. Thus, the thermal budget of fabricating the memory cell array does not affect the fabrication of the peripheral circuits. The stacked memory cell array and peripheral circuits can also reduce the chip size compared with the side-by-side arrangement, thereby improving the array efficiency. In some implementations, more than one memory cell array is stacked over one another using bonding techniques to further increase the array efficiency. In some implementations, the word lines and bit lines are disposed close to the bonding interface due to the vertically arranged transistors, which can be coupled to the peripheral circuits through a large number (e.g., millions) of parallel bonding contacts across the bonding interface can make direct, short-distance (e.g., micron-level) electrical connections between the memory cell array and peripheral circuits to increase the throughput and input/output (I/O) speed of the memory devices.
[0031] In some implementations, the vertical transistors disclosed herein include multi-gate transistors (e.g., gate-all-around (GAA) transistors, tri-gate transistors, or double-gate transistors), which can have a larger gate control area to achieve better channel control with a smaller subthreshold swing. Since the channel is fully depleted, the leakage current of multi-gate transistors can be significantly reduced as well. Thus, using multi-gate transistors instead of planar transistors can achieve a much better speed (saturated drain current)/leakage current performance.
[0032] In some implementations, the vertical transistors disclosed herein include single-gate transistors (a.k.a. single-side gate transistors) in a mirror-symmetric arrangement with respect to adjacent transistors in the bit line direction as a result of splitting multi-gate transistors (e.g., double-gate transistors) using trench isolations extending along the word line direction. Thus, the memory cell density in the bit line direction can be significantly increased (e.g., doubled) without unduly complicating the fabrication process compared with using processes, such as self-aligned double patterning (SADP). Also, the mirror-symmetric single-gate transistors have a larger process window for word line, bit line, and transistor pitch reduction, compared to either planar transistors or multi-gate vertical transistors, for example, with dual-side or all-around gates.
[0033] FIG. **1** illustrates a block diagram of system **100** having a memory device, according to some aspects of the present disclosure. System **100** can be a mobile phone, a desktop computer, a laptop computer, a tablet, a vehicle computer, a gaming console, a printer, a positioning device, a wearable electronic device, a smart sensor, a virtual reality (VR) device, an argument reality (AR) device, or any other suitable electronic devices having storage therein. As shown in FIG. **1**, system **100** can include a host **108** and a memory system **102** having one or more memory devices **104** and a memory controller **106**. Host **108** can be a processor of an electronic device, such as a central processing unit (CPU), or a system-on-chip (SoC), such as an application processor (AP). Host **108** can be configured to send or receive the data to or from memory devices **104**.
[0034] Memory device **104** can be any memory device disclosed herein. In some implementations, memory device **104** includes an array of memory cells each including a vertical transistor, as described herein.
[0035] Memory controller **106** is coupled to memory device **104** and host **108** and is configured to control memory device **104**, according to some implementations. Memory controller **106** can manage the data stored in memory device **104** and communicate with host **108**. Memory controller **106** can be configured to control operations of memory device **104**, such as read, write, and refresh operations. Memory controller **106** can also be configured to manage various functions with respect to the data stored or to be stored in memory device **104** including, but not limited to refresh and timing control, command/request translation, buffer and schedule, and power management. In some implementations, memory controller **106** is further configured to determine the maximum memory capacity that the computer system can use, the number of memory banks, memory type and speed, memory particle data depth and data width, and other important parameters.
[0036] Any other suitable functions may be performed by memory controller **106** as well. Memory

controller **106** can communicate with an external device (e.g., host **108**) according to a particular communication protocol. For example, memory controller **106** may communicate with the external device through at least one of various interface protocols, such as a USB protocol, an MMC protocol, a peripheral component interconnection (PCI) protocol, a PCI-express (PCI-E) protocol, an advanced technology attachment (ATA) protocol, a serial-ATA protocol, a parallel-ATA protocol, a small computer small interface (SCSI) protocol, an enhanced small disk interface (ESDI) protocol, an integrated drive electronics (IDE) protocol, a Firewire protocol, etc.

[0037] FIG. **2** illustrates a schematic diagram of a memory device **200** including peripheral circuits and an array of memory cells each having a vertical transistor, according to some aspects of the present disclosure. Memory device **200** can include a memory cell array **201** and peripheral circuits **202** coupled to memory cell array **201**. Memory cell array **201** can be any suitable memory cell array in which each memory cell **208** includes a vertical transistor **210** and a storage unit **212** coupled to vertical transistor **210**. In some implementations, memory cell array **201** is a DRAM cell array, and storage unit **212** is a capacitor for storing charge as the binary information stored by the respective DRAM cell. In some implementations, memory cell array **201** is a PCM cell array, and storage unit **212** is a PCM element (e.g., including chalcogenide alloys) for storing binary information of the respective PCM cell based on the different resistivities of the PCM element in the amorphous phase and the crystalline phase. In some implementations, memory cell array **201** is a FRAM cell array, and storage unit **212** is a ferroelectric capacitor for storing binary information of the respective FRAM cell based on the switch between two polarization states of ferroelectric materials under an external electric field.

[0038] As shown in FIG. **2**, memory cells **208** can be arranged in a two-dimensional (2D) array having rows and columns. Memory device **200** can include word lines **204** coupling peripheral circuits **202** and memory cell array **201** for controlling the switch of vertical transistors **210** in memory cells **208** located in a row, as well as bit lines **206** coupling peripheral circuits **202** and memory cell array **201** for sending data to and/or receiving data from memory cells **208** located in a column. That is, each word line **204** is coupled to a respective row of memory cells **208**, and each bit line is coupled to a respective column of memory cells **208**.

[0039] Vertical transistors **210**, such as vertical metal-oxide-semiconductor field-effect transistors (MOSFETs), can replace the planar transistors as the pass transistors of memory cells **208** to reduce the area occupied by the pass transistors, the coupling capacitance, as well as the interconnect routing complexity. As shown in FIG. **2**, in some implementations, different from planar transistors in which the active regions are formed in the substrates, vertical transistor **210** includes a semiconductor body **214** extending vertically (in the z-direction) above the substrate (not shown). That is, semiconductor body **214** can extend above the top surface of the substrate to allow channels to be formed not only at the top surface of semiconductor body **214**, but also at one or more side surfaces thereof.

[0040] As shown in FIG. **2**, for example, semiconductor body **214** can have a cuboid shape to expose four sides thereof. It is understood that semiconductor body **214** may have any suitable 3D shape, such as a polyhedron shape or a cylinder shape. That is, the cross-section of semiconductor body **214** in the plan view (e.g., in the x-y plane) can have a square shape, a rectangular shape (or a trapezoidal shape), a circular (or an oval shape), or any other suitable shapes. It is understood that, for semiconductor bodies that have a circular or oval shape of their cross-sections in the plan view, the semiconductor bodies may still be considered as having multiple sides, such that the gate structures are in contact with more than one side of the semiconductor bodies. As described below with respect to the fabrication process, in some cases, semiconductor body **214** can be formed from the substrate (e.g., by etching or epitaxy) and thus, has the same semiconductor material (e.g., silicon crystalline silicon) as the substrate (e.g., a silicon substrate).

[0041] As shown in FIG. **2** example, vertical transistor **210** can also include a gate structure **216** in contact with one or more sides of semiconductor body **214**, e.g., in one or more planes of the side

surface(s) of the active region. In other words, the active region of vertical transistor **210**, e.g., semiconductor body **214**, can be at least partially surrounded by gate structure **216**. Gate structure **216** can include a gate dielectric **218** over one or more sides of semiconductor body **214**, e.g., in contact with four side surfaces of semiconductor body **214** as shown in FIG. **2**. Gate structure **216** can also include a gate electrode **220** over and in contact with gate dielectric **218**. Gate dielectric **218** can include any suitable dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or high-k dielectrics. For example, gate dielectric **218** may include silicon oxide, which is a form of gate oxide. Gate electrode **220** can include any suitable conductive materials, such as polysilicon, metals (e.g., tungsten (W), copper (Cu), aluminum (Al), etc.), metal compounds (e.g., titanium nitride (TiN), tantalum nitride (TaN), etc.), or silicides. For example, gate electrode **220** may include doped polysilicon, which is a form of a gate poly. In some implementations, gate electrode **220** includes multiple conductive layers, such as a W layer over a TiN layer. It is understood that gate electrode **220** and word line **204** may be a continuous conductive structure in some examples. In other words, gate electrode **220** may be viewed as part of word line **204** that forms gate structure **216**, or word line **204** may be viewed as the extension of gate electrode **220** to be coupled to peripheral circuits **202**.
[0042] As shown in FIG. **2**, vertical transistor **210** can further include a pair of a source and a drain (S/D, dope regions, a.k.a., source electrode and drain electrode) formed at the two ends of semiconductor body **214** in the vertical direction (the z-direction), respectively. The source and drain can be doped with any suitable P-type dopants, such as boron (B) or Gallium (Ga), or any suitable N-type dopants, such as phosphorus (P) or arsenic (As). The source and drain can be separated by gate structure **216** in the vertical direction (the z-direction). In other words, gate structure **216** is formed vertically between the source and drain. As a result, one or more channels (not shown) of vertical transistor **210** can be formed in semiconductor body **214** vertically between the source and drain when a gate voltage applied to gate electrode **220** of gate structure **216** is above the threshold voltage of vertical transistor **210**. That is, each channel of vertical transistors **210** is also formed in the vertical direction along which semiconductor body **214** extends, according to some implementations.
[0043] In some implementations, as shown in FIG. **2**, vertical transistor **210** is a multi-gate transistor. That is, gate structure **216** can be in contact with more than one side of semiconductor body **214** (e.g., four sides in FIG. **2**) to form more than one gate, such that more than one channel can be formed between the source and drain in operation. That is, different from the planar transistor that includes only a single planar gate (and resulting in a single planar channel), vertical transistor **210** shown in FIG. **2** can include multiple vertical gates on multiple sides of semiconductor body **214** due to the **3**D structure of semiconductor body **214** and gate structure **216** that surrounds the multiple sides of semiconductor body **214**. As a result, compared with planar transistors, vertical transistor **210** shown in FIG. **2** can have a larger gate control area to achieve better channel control with a smaller subthreshold swing. Since the channel is fully depleted, the leakage current (Ioff) of vertical transistor **210** can be significantly reduced a well. In various examples, the multi-gate vertical transistors can include double-gate vertical transistors (e.g., dual-side gate vertical transistors), tri-gate vertical transistors (e.g., tri-side gate vertical transistors), and GAA vertical transistors.
[0044] It is understood that although vertical transistor **210** is shown as a multi-gate transistor in FIG. **2**, the vertical transistors disclosed herein may also include single-gate transistors. That is, gate structure **216** may be in contact with a single side of semiconductor body **214**, for example, for the purpose of increasing the transistor and memory cell density. It is also understood that although gate dielectric **218** is shown as being separate (a separate structure) from other gate dielectrics of adjacent vertical transistors (not shown), gate dielectric **218** may be part of a continuous dielectric layer having multiple gate dielectrics of vertical transistors.
[0045] In planar transistors and some lateral multiple-gate transistors (e.g., FinFET), the active

regions, such as semiconductor bodies (e.g., Fins), extend laterally (in the x-y plane), and the source and the drain are disposed at different locations in the same lateral plane (the x-y plane). In contrast, in vertical transistor **210**, semiconductor body **214** extends vertically (in the z-direction), and the source and the drain are disposed in the different lateral planes, according to some implementations. In some implementations, the source and the drain are formed at two ends of the semiconductor body **214** in the vertical direction (the z-direction), respectively, thereby being overlapped in the plan view. As a result, the area (in the x-y plane) occupied by vertical transistor **210** can be reduced compared with planar transistors and lateral multiple-gate transistors. Also, the metal wiring coupled to vertical transistors **210** can be simplified as well since the interconnects can be routed in different planes. For example, bit lines **206** and storage units **212** may be formed on opposite sides of vertical transistor **210**. In one example, bit line **206** may be coupled to the source or the drain at the upper end of semiconductor body **214**, while storage unit **212** may be coupled to the other source or the drain at the lower end of semiconductor body **214**.

[0046] As shown in FIG. **2**, storage unit **212** can be coupled to the source or the drain of vertical transistor **210**. Storage unit **212** can include any devices that are capable of storing binary data (e.g., 0 and 1), including but not limited to, capacitors for DRAM cells and FRAM cells, and PCM elements for PCM cells. In some implementations, vertical transistor **210** controls the selection and/or the state switch of the respective storage unit **212** coupled to vertical transistor **210**. In some implementations, as shown in FIG. **3**, each memory cell **208** is a DRAM cell **302** including a transistor **304** (e.g., implementing using vertical transistors **210** in FIG. **2**) and a capacitor **306** (e.g., an example of storage unit **212** in FIG. **2**). The gate of transistor **304** (e.g., corresponding to gate electrode **220**) may be coupled to word line **204**, one of the source and the drain of transistor **304** may be coupled to bit line **206**, the other one of the source and the drain of transistor **304** may be coupled to one electrode of capacitor **306**, and the other electrode of capacitor **306** may be coupled to the ground.

[0047] In some implementations, as shown in FIG. **4**, each memory cell **208** is a PCM cell **402** including a transistor **404** (e.g., implementing using vertical transistors **210** in FIG. **2**) and a PCM clement **406** (e.g., an example of storage unit **212** in FIG. **2**). The gate of transistor **404** (e.g., corresponding to gate electrode **220**) may be coupled to word line **204**, one of the source and the drain of transistor **404** may be coupled to the ground, the other one of the source and the drain of transistor **404** may be coupled to one electrode of PCM element **406**, and the other electrode of PCM element **406** may be coupled to bit line **206**.

[0048] Peripheral circuits **202** can be coupled to memory cell array **201** through bit lines **206**, word lines **204**, and any other suitable metal wirings. As described above, peripheral circuits **202** can include any suitable circuits for facilitating the operations of memory cell array **201** by applying and sensing voltage signals and/or current signals through word lines **204** and bit lines **206** to and from each memory cell **208**. For example, peripheral circuits **202** can include various types of peripheral circuits formed using CMOS technologies.

II. Fabrication Process to Protect Word Line in Backside Process

[0049] FIGS. **5**A-**5**B illustrate a result of a fabrication process where trench recess regions overlap insulating layers for bit line formation at a backside of a semiconductor device. FIG. **5**A illustrates the top-view of the overlapped trench recess regions and insulating layers. FIG. **5**B illustrates a cross-section view of the semiconductor device along the word line direction. During the fabrication process, at the backside of the semiconductor device, the trench recess region **501** can be defined to recess the trench isolations **504** and gate structures (word lines **516**) along the bit line direction. A spacer **505** can be deposited to cover the surface of the space resulting from the recess. The space can later be filled with a sacrificial material **506**. The spacer **505** can be any suitable dielectric materials, such as silicon nitride, silicon oxide, silicon oxynitride, or high-k dielectrics. The sacrificial material **506** can be any suitable conductive or non-conductive materials, such as polysilicon, metals (e.g., tungsten (W), copper (Cu), aluminum (Al), etc.), metal compounds (e.g., titanium nitride (TiN), tantalum nitride (TaN), etc.), or silicides.

[0050] Insulating layer **502** can then be formed at the backside of the semiconductor device. As shown in FIG. **5**A and **5**B, the insulating layer **502** covers a portion of the trench recess region **501** resulting in an overlap between the insulating layer **502** and the trench recess region **501**. The sacrificial materials **506** that are not covered by the insulating layer **502** are removed by performing an etch process to form air gaps between semiconductor bodies **514**. Bit lines **518** can then be formed on top of the semiconductor bodies **514** along the bit line direction in the trench recess region **501** that are not covered by the insulating layer **502**. However, due to the insulating layer **502**, the sacrificial materials **506** covered by the insulating layer **502** are not removed. The existing of the sacrificial materials **506** can potentially cause current leakage. It is desirable to completely remove the sacrificial materials **506** under the insulating layer **502**.

[0051] FIGS. **6**A-**6**B illustrate a result of a fabrication process where trench recess regions and insulating layers for bit line formation are not overlapped with each other at a backside of the semiconductor device. FIG. **6**A illustrates the top-view of the non-overlapped trench recess regions and insulating layers. FIG. **6**B illustrates a cross-section view of the semiconductor device along the word line direction. At the backside of the semiconductor device, the trench recess region **601** can be defined to recess the trench isolations **604** and gate structures **616** along the bit line direction. Similarly, after deposition of a spacer **605**, the space resulting from the recess can then be filled with a sacrificial material **606**. The spacer **605** can be any suitable dielectric materials, such as silicon nitride, silicon oxide, silicon oxynitride, or high-k dielectrics. The sacrificial material **606** can be any suitable conductive materials, such as polysilicon, metals (e.g., tungsten (W), copper (Cu), aluminum (Al), etc.), metal compounds (e.g., titanium nitride (TiN), tantalum nitride (TaN), etc.), or silicides.

[0052] Insulating layer **602** can then be formed at the backside of the semiconductor device. As shown in FIG. **6**A and **6**B, the insulating layer **602** and the trench recess region **601** do not overlap, resulting in a few trench isolations **604** and semiconductor bodies **614** not covered by any of the insulating layer **602** and trench recess region **601**. The sacrificial materials **606** are removed by performing an etch process, to form air gaps between semiconductor bodies **614**. Bit lines **618** can then be formed on semiconductor bodies **614** along the bit line direction that are not covered by the insulating layer **602**. Due to the difficulty of controlling a uniform depth of trenches for the gate structures in the semiconductor device with a vertical all in one etch process, certain gate structures **616** may have abnormal heights when viewed from the backside of the semiconductor device. During the removal process of the sacrificial materials **606**, gate structures **616** with abnormal heights that are not covered by the trench recess regions **601** and not covered by the insulating layers **602** may also be removed together from the backside of the semiconductor device, which may result in word line missing (or broken) as shown in FIG. **6**B. To mitigate this word line broken issue, the current disclosure provides a backside processing method where the insulating layer **602** is disposed to overlap the trench recess region **601**, reducing or eliminating the chance of accidentally etching the word lines. At the same, a tunnel structure crossing below bit lines is employed to remove the sacrificial materials **506** under the insulating layer **502** (as shown in FIG. **5**B), which can solve the leakage issue.

[0053] FIGS. **7**A-**1**˜**7**A-**2**/**7**B-**1**˜**7**B-**2**/**7**C-**1**˜**7**C-**2**/**7**D-**1**˜**7**D-**5**/**7**E/**7**F-**1**˜**7**F-**5**/**7**G-**1**˜**7**G-**7**/**7**H-**1**˜**7**H-**7**/**7**I-**1**˜**7**I-**7**/**7**J-**1**˜**7**J-**7**/**7**K-**1**˜**7**K-**7** illustrate a fabrication process for forming a memory device according to some aspects of the present disclosure.

[0054] FIG. **7**A-**1** illustrates a plan view of a semiconductor memory device **700** having four sections from a topside **701**. FIGS. **7**A-**2** illustrates a partial section view corresponding to the cut line X**1**-X**1**. In FIGS. **7**A-**1**/**7**A-**2**, trench isolations **704** (e.g., STIs) are formed in a semiconductor substrate **702** of a semiconductor memory device **700** in a bit line direction (e.g., y direction). The semiconductor substrate **702** can be a silicon substrate. The semiconductor memory device **700** (as well as the semiconductor substrate **702**) can have the topside **701** and a backside **703**. The trench isolations **704** are formed from the topside **701**. In some implementations, a lithography process is performed to pattern trenches and semiconductor walls **705** in the semiconductor substrate **702** using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of bit lines. One or more dry etching and/or wet etching processes, such as reactive ion etch (RIE), are performed on the semiconductor substrate **702**. In some implementations, a dielectric, such as silicon oxide, is deposited to fully fill the trenches to form trench isolations **704** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, a planarization process, such as CMP, is performed to remove excess dielectric deposited beyond the top surface of the topside **701**.

[0055] FIG. 7B-**1** illustrates a plan view of the semiconductor memory device **700** having four sections from the topside **701**. FIGS. 7B-**2** illustrates a partial section view corresponding to the cut line Y**1**-Y**1**. In FIGS. 7B-**1**/7B-**2**, word line trenches **706** are formed along the word line direction (e.g., x direction). In some implementations, a lithography process is performed to pattern word line trenches **706** to be perpendicular to trench isolations **704** using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of word lines. One or more dry etching and/or wet etching processes, such as RIE, are performed on semiconductor wall **705** and trench isolations **704** to etch word line trenches **706** in semiconductor substrate **702**. As a result, semiconductor bodies **708** are formed as the result of forming the word line trenches **706** and the trench isolations **704**. The semiconductor bodies **708** can extend in a vertical direction (e.g., z direction). Word line layers **711** can be formed in the word line trenches **706** along the sidewalls of semiconductor bodies **708** and connect to the sidewalls of semiconductor bodies **708** in a row. The word line layers **711** includes word lines and gate structures (not shown) of the semiconductor bodies **708** can be formed in a similar manner as described in FIG. **2** above. At this stage, the word line layers **711** are connected at the bottom of each corresponding word line trench **706**. Word lines connect rows of the semiconductor bodies **708** are formed in a later process discussed below. Metal shields **710** can be formed in the word line trenches **706** and between two neighboring word line layers **711** in a similar process as forming the word lines. The word line trenches **706** can then be filled with an insulating material, such as silicon oxide.

[0056] FIG. 7C-**1** illustrates a plan view of the semiconductor memory device **700** having four sections from the topside **701**. FIGS. 7C-**2** illustrates a partial section view corresponding to the cut line Y**1**-Y**1**. In FIGS. 7C-**1**/7C-**2**, portions **714** of word line layers **711** are recessed down from the topside **701** at desired locations. Each word line layer **711** has two portions **714** being recessed down to form two separated word lines that disconnect at the recessed region. That is, each row of the semiconductor body **708** along the word line direction (x direction) is in contact with only one word line. At this stage, the word line layer **711** within each word line trench **704** includes two separated word lines that are still connected at the bottom of each word line trench **704**. The bottom connected portion will be disconnected in a backside process detailed below. In some implementations, a lithography process is performed to pattern the portions **714** using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of word lines. One or more dry etching and/or wet etching processes, such as reactive ion etch (RIE), can be performed to etch and recess the word line layers **711**. In some implementations, the portions **714** of word line layers **711** being recessed are located close to ends of each word line trenches **706** and within trench isolations **704** to maximize the number of semiconductor bodies **708** connected in one row by one word line. The portions **714** can then be filled with insulating materials, such as silicon nitride.

[0057] FIG. **7D-1** illustrates an enlarged plan view of one of the sections of the semiconductor memory device **700** from the backside **703**. FIGS. **7D-2** and **7D-3** illustrate partial section views corresponding to cut line Y**1**-Y**1** and cut line Y**2**-Y**2** respectively. FIGS. **7D-4** and **7D-5** illustrate partial section views corresponding to cut line Y**1**-Y**1** and cut line Y**2**-Y**2** respectively. In FIGS. **7D-1**/**7D-2**/**7D-3**/**7D-4**/**7D-5**, the semiconductor substrate **702** is thinned from the backside **703** after storage units **720** are formed on respective semiconductor bodies **708**. Upper ends of the semiconductor bodies **708** at the topside **701** can be doped. The exposed end of each semiconductor body **708** is doped to form a source/drain (e.g., a source terminal of a vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **708** to form sources/drains. In some implementations, a silicide layer is formed on source/drain by performing a silicidation process at the exposed upper ends of semiconductor bodies **708**.

[0058] Storage units in contact with the semiconductor bodies, e.g., the doped first ends thereof, are formed. The storage unit can include a capacitor or a PCM element. In some implementations, to form a storage unit that is a capacitor, a first electrode is formed on the doped upper end of the semiconductor body, a capacitor dielectric is formed on the first electrode, and a second electrode is formed on the capacitor dielectric.

[0059] For example, one or more interlayer dielectric (ILD) layers **716** are formed over the topside **701**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Storage unit contact (capacitor contact) **718**, first electrodes, capacitor dielectrics, and second electrodes of storage units (capacitors) **720**, and a common plate **722** are subsequently formed in the ILD layers **716** to be coupled to semiconductor bodies **708**. In some implementations, capacitor contact **718** is formed on a respective source/drain, e.g., the doped upper end of a respective semiconductor body **708** by patterning and etching an electrode hole aligned with the respective source/drain using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementations, common plate **722** is formed on the second electrodes of capacitors **720** by patterning and etching an electrode trench aligned with capacitors **720** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

[0060] The semiconductor substrate **702** can be thinned at the backside **703** using CMP, grinding, dry etching, and/or wet etching. In some embodiments, the CMP process is performed to thin the semiconductor substrate **702** until reaching the trench isolations **704**. After thinning the backside **703**, trench isolations **704** are exposed. Lines of semiconductor structures **707** in the bit line direction (y direction) are exposed and connect columns of the semiconductor bodies **708** at the backside **703**. The line of semiconductor structure **707** along with the respective columns of connected semiconductor bodies **708** forms a bridge-shaped structure, where the line of the semiconductor structure **707** is the beam and connected semiconductor bodies **708** are the piers. As illustrated in FIG. 7D-**4**, due to the difficulty in controlling a uniform depth of gate structures of the word line **712** with vertical all in one etch, the word line **712** may have abnormally higher regions **713** in the trench isolations **704**.

[0061] FIG. **7**E illustrates an enlarged plan view of one of the sections of the semiconductor memory device **700** from the backside **703**. Trench recess regions **727** can be patterned out by applying a mask **723** at the backside **703**. The locations of the trench recess regions **727** can be determined according to the functionalities of the specific design of the semiconductor memory device.

[0062] FIG. **7**F-**1** illustrates an enlarged plan view of one of the sections of the semiconductor memory device **700** from the backside **703**. FIGS. **7**F-**2** and **7**F-**3** illustrate partial section views corresponding to cut line Y**1**-Y**1** and cut line Y**2**-Y**2** respectively. FIGS. **7**F-**4** and **7**F-**5** illustrate partial section views corresponding to cut line Y**1**-Y**1** and cut line Y**2**-Y**2** respectively. In FIGS. **7**F-**1**/**7**F-**2**/**7**F-**3**/**7**F-**4**/**7**F-**5**, the trench isolations **704** within the trench recess regions **727** are recessed to expose the word line layers **711** and the metal shields **710**. The word line layers **711** at the bottom of the word line trenches **706** are etched away to form two separated word lines **712** that each connects one row of semiconductor bodies **708**. The word lines **712** and the metal shields **710** are further recessed down towards the topside **701** within the trench recess regions **727**. FIG. **7**F-**4** illustrates the word line **712** that is further recessed towards the topside **701** within the trench recess region **727**. FIG. **7**F-**5** illustrates the metal shield **710** that is further recessed towards the topside **701** within the trench recess region **727**. One or more dry etching and/or wet etching processes, such as reactive ion etch (RIE), can be performed to etch and recess the trench isolations **704**, word lines **712**, and metal shields **710** within the trench recess regions **727**.

[0063] Tunnels **724** are formed underneath the bridge-shaped structures along the bit line direction (y direction) within the trench recess regions **727**. Tunnels **724** can have different shapes according to the respective word line **712** or metal shield **710**. For example, as shown in FIG. **7**F-**2**, tunnels **724** connected with the word lines **712** have a “P” shape, and tunnels **724** connected with the metal shields **710** have a “T” shape, as viewed at the Y**1**-Y**1** cut line. For example, as shown in FIG. **7**F-**3**, tunnels **724** connected with the word lines **712** have an “n” shape, as viewed at the Y**2**-Y**2** cut line.

[0064] In some implementations, a lithography process can be performed to pattern the trench recess regions **727** by using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of bit lines. One or more dry etching and/or wet etching processes, such as RIE, can be performed on the semiconductor substrate **702** to etch away the trench isolations **704**. After the trench isolations **704** are recessed, a second lithography process can be performed to pattern the word lines **712** and metal shields **710** within the trench recess regions **727**. One or more dry etching and/or wet etching processes, such as RIE, can be performed to etch away the word lines **712** and metal shields **710** within the trench recess regions **727**.

[0065] FIG. **7**G-**1** illustrates an enlarged plan view of one of the sections of the semiconductor memory device **700** from the backside **703**. FIGS. **7**G-**2** and **7**G-**3** illustrate partial section views corresponding to the cut line Y**1**-Y**1** and cut line Y**2**-Y**2** respectively. FIG. **7**G-**4** illustrates a partial section view between cut lines Y**1**-Y**1** and Y**3**-Y**3**. FIG. **7**G-**5** illustrates a partial section view between cut lines Y**2**-Y**2** and Y**4**-Y**4**. FIGS. **7**F-**6** and **7**F-**7** illustrate partial section views corresponding to cut line Y**1**-Y**1** and cut line Y**2**-Y**2** respectively. In FIGS. **7**G-**1**/**7**G-**2**/**7**G-**3**/**7**G-**4**/**7**G-**5**/**7**G-**6**/**7**G-**7**, a thin spacer layer **726** is deposited on the sidewalls **725** of the tunnels **724** and recessed trench isolations **704**. As shown in FIGS. **7**G-**4** and **7**G-**5**, the thin spacer layer **726** only formed on the sidewalls **725** of tunnels **724** and does not fill the tunnels **724** and the trench isolations **704**. The thin spacer layer **726** is formed by depositing a layer of dielectric materials, such as silicon nitride, by using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without fully filling the trench isolations **704** and the tunnels **724**.

[0066] FIG. **7**H-**1** illustrates an enlarged plan view of one of the sections of the semiconductor memory device **700** from the backside **703**. FIGS. **7**H-**2** and **7**H-**3** illustrate partial section views corresponding to the cut line Y**1**-Y**1** and cut line Y**2**-Y**2** respectively. FIGS. **7**H-**4**, **7**H-**5**, **7**H-**6**, and **7**H-**7** illustrate partial section views corresponding to cut line X**1**-X**1**, cut line X**2**-X**2**, cut line X**3**-X**3**, and cut line X**4**-X**4** respectively. In FIGS. **7**H-**1**/**7**H-**2**/**7**H-**3**/**7**H-**4**/**7**H-**5**/**7**H-**6**/**7**H-**7**, a sacrificial layer **728** is deposited on the semiconductor substrate **702** from the backside **703**. The sacrificial layer **728** is fully filled in the trench isolations **704** and the tunnels **724**. The sacrificial layer **728** can be any suitable sacrificial materials that can be later selectively removed, such as metals (e.g., tungsten (W), copper (Cu), aluminum (Al), etc.), or metal compounds (e.g., titanium nitride (TiN), tantalum nitride (TaN), etc.). The sacrificial layer **728** can be deposited using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof to fully fill the trench isolations **704** and the tunnels **724**.

[0067] FIG. **7**I-**1** illustrates an enlarged plan view of one of the sections of the semiconductor memory device **700** from the backside **703**. FIGS. **7**I-**2** and **7**I-**3** illustrate partial section views corresponding to the cut line Y**1**-Y**1** and cut line Y**2**-Y**2** respectively. FIGS. **7**I-**4**, **7**I-**5**, **7**I-**6**, and **7**I-**7** illustrate partial section views corresponding to cut line X**1**-X**1**, cut line X**2**-X**2**, cut line X**3**-X**3**, and cut line X**4**-X**4** respectively. In FIGS. **7**I-**1**/**7**I-**2**/**7**I-**3**/**7**I-**4**/**7**I-**5**/**7**I-**6**/**7**I-**7**, excessive sacrificial material of the sacrificial layer **728**, and excessive dielectric materials of the thin spacer layer **726** are removed from the surface of the backside **703**. In some implementations, a planarization process, such as CMP, is performed to remove the excessive material.

[0068] FIG. **7J-1** illustrates an enlarged plan view of one of the sections of the semiconductor memory device **700** from the backside **703**. FIGS. **7J-2** and **7J-3** illustrate partial section views corresponding to the cut line Y**1**-Y**1** and cut line Y**2**-Y**2** respectively. FIGS. **7J-4**, **7J-5**, **7J-6**, and **7J-7** illustrate partial section views corresponding to cut line X**1**-X**1**, cut line X**2**-X**2**, cut line X**3**-X**3**, and cut line X**4**-X**4** respectively. In FIGS. **7J-1**/**7J-2**/**7J-3**/**7J-4**/**7J-5**/**7J-6**/**7J-7**, an insulating layer **730** is formed on the backside **703** of the semiconductor substrate **702**. The insulating layer **730** selectively covers the semiconductor structures **707** and trench isolations **704** within the trench recess region **727**. That is, the insulating layer **730** overlaps with the trench recess regions **727**. In other words, portions **731** of the backside **703** that are not covered by the insulating layer **730** are smaller than the respective trench recess region **727**. As shown in FIGS. **7J-4** and **7J-5**, the insulating layer **730** also covers the abnormally higher regions **713** of the word line **712**. This way, the word lines **712** are protected from any accidental etching. Portions **731** not covered by the insulating layer **730** are selected as where the bit line of the semiconductor memory device **700** will be formed in a later process. In some implementations, the insulating layer **730** can include any suitable insulating materials, such as silicon oxide, to be deposited on the backside **703** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, a planarization process, such as CMP, is performed to remove excess insulating material. A lithography process can be performed to pattern the insulating layer **730** to selectively cover the backside **703** and have portions **731** not covered by using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of bit lines. One or more dry etching and/or wet etching processes, such as RIE, can be performed on the insulating layer **730** to etch away portions **731**.

[0069] FIG. **7K-1** illustrates an enlarged plan view of one of the sections of the semiconductor memory device **700** from the backside **703**. FIG. **7K-2** illustrates partial section views corresponding to the cut line Y**1**-Y**1**. FIG. **7K-3** illustrates partial section views between cut lines Y**2**-Y**2** and Y**4**-Y**4**. FIGS. **7J-4** and **7J-6** illustrate partial section views corresponding to cut line X**1**-X**1** and cut line X**3**-X**3** respectively. FIG. **7K-5** illustrates partial section views between cut lines X**1**-X**1** and X**2**-X**2**. FIG. **7K-7** illustrates partial section views between cut lines X**3**-X**3** and X**4**-X**4**. In FIGS. **7K-1**/**7K-2**/**7K-3**/**7K-4**/**7K-5**/**7K-6**/**7K-7**, bit lines **732** are formed on the exposed semiconductor structures **707** followed by the removal of the sacrificial layer **728**. The semiconductor structures **707** exposed in the portions **730** can be doped with N-type dopants (e.g., P or As) or P-type dopants (e.g., B or Ga) at a desired doping level, by performing an implantation process, a thermal diffusion process, a combination thereof, or any suitable processes. In some implementations, the backside **703** of the semiconductor substrate **702** can be pre-doped in an earlier process. The doping discussed above can be used to further adjust the doping level of the semiconductor structures **707**. In some implementations, the doping operation can be skipped.

[0070] A metal layer can be deposited on top of the semiconductor structures **707** to fill the portions **731**. The metal layer can include W, Co, Cu, Al, or any other suitable metals having higher conductivities than the doped substrate. A high-temperature annealing process can be performed on the semiconductor substrate **702** (e.g., a silicon substrate) to form metal silicide (i.e., bit lines **732**) on semiconductor structures **707**. Metal atoms in the metal layer are diffused into the underlying semiconductor structures **707** (i.e., the silicon substrate) to form the metal silicide (i.e., bit lines **732**). The metal atoms would not react when in contact with the sacrificial layer **728** as the sacrificial material is selected to be metal or metal compound as described in FIG. **7H-1**.

[0071] A metal removal process can be performed to remove any excess metal layer and the sacrificial layer **728**. One or more dry etching and/or wet etching processes, such as RIE, can be performed to etch away the remaining metal layer and the sacrificial layer **728**. As shown in FIGS.

7K-5 and **7K-7**, the sacrificial layer **728** covered by the insulating layer **730** below the covered respective semiconductor structures **707** can be removed by allowing the selective etchant flow through tunnels **724**. The abnormally higher regions **713** of the word line **712** are protected by the insulating layer **730** while all sacrificial material of the sacrificial layer **728** are etched away.

[0072] By utilizing an overlap between the trench recess region **727** and the insulating layer **730** while introducing a tunnel structure (i.e., tunnels **724**), all sacrificial material can be removed which avoids current leakage from any non-removed sacrificial material. In addition, the abnormally higher regions **713** of the word line **712** are protected by the insulating layer **730** from being accidentally etched away during the removal of the sacrificial layer **728** to prevent the word line missing issue.

III. Flowchart of Fabrication Process to Protect Word Line in Backside Process

[0073] FIG. **8** illustrates a flowchart of a fabrication process **800** for forming a 3D memory device including vertical transistors, according to some aspects of the present disclosure. The fabrication process **800** is just an example of various methods and techniques disclosed in this disclosure. Not all steps are performed in various embodiments. The steps can be carried out in different orders or in parallel in various embodiments. The fabrication process **800** starts from S**801** and proceeds to S**810**.

[0074] At S**810**, a semiconductor memory device can be thinned from a backside of the semiconductor device to expose trench isolations. The semiconductor memory device can have the trench isolations formed in a bit-line direction, gate structures formed in a word-line direction perpendicular to the bit-line direction, and an array of vertical-transistor channel structures that extend in a vertical direction perpendicular to the bit-line direction and the word-line direction and are separated by the trench isolations and the gate structures. After the thinning, the trench isolations are exposed at the back of the semiconductor device. Top ends of the vertical-transistor channel structures in each column along the bit-line direction are connected with a line of semiconductor structure extending along the bit-line direction at the backside of the semiconductor memory device to form a bridge-shaped structure. The trench isolations separate neighboring bridge-shaped structures. The semiconductor memory device can further have metal shields formed between every two neighboring channel structures along the word-line direction.

[0075] At S**820**, the trench isolations between the neighboring bridge-shaped structures can be recessed in a first region of the backside of the semiconductor memory device to expose the gate structures formed in the word-line direction.

[0076] At S**830**, the gate structures formed in the word-line direction can be etched, resulting in tunnels along the respective gate structures. The tunnels cross below each line of semiconductor structures.

[0077] At S**840**, surfaces of a space where the trench isolations and the gate structures are etched away can be covered with a spacer layer.

[0078] At S**850**, the space where the trench isolations and the gate structures are etched away can be filled with a sacrificial material over the spacer layer.

[0079] At S**860**, a portion of the first region of the backside of the semiconductor memory device can be covered with an insulation layer. The insulation layer can enclose the sacrificial materials below the insulation layer and between the respective neighboring bridge-shaped structures at the covered portion of the first region. The insulation layer can cover a second region neighboring the portion of the first region covered by the insulation layer, and the trench isolations in the second region are maintained while the trench isolations in the first region are etched.

[0080] At S**870**, bit-line structures can be formed over the lines of semiconductor structures not covered by the insulation layer in the first region while the portion of the first region is covered with the insulation layer.

[0081] At S**880**, the sacrificial material within the space where the trench isolations and the gate structures are etched away can be removed while the portion of the first region is covered with the insulation layer. The sacrificial material enclosed by the insulation layer is removed through the tunnels crossing below the respective lines of semiconductor structures. The process proceeds to S**899** and terminates at S**899**.

[0082] While aspects of the present disclosure have been described in conjunction with the specific embodiments thereof that are proposed as examples, alternatives, modifications, and variations to the examples may be made. Accordingly, embodiments as set forth herein are intended to be illustrative and not limiting. There are changes that may be made without departing from the scope of the claims set forth below.

Claims

1. A method of fabricating a semiconductor memory device, comprising: thinning the semiconductor memory device from a backside of the semiconductor device that, at current stage, has trench isolations formed in a bit-line direction, gate structures formed in a word-line direction perpendicular to the bit-line direction, and an array of vertical-transistor channel structures that extend in a vertical direction perpendicular to the bit-line direction and the word-line direction and are separated by the trench isolations and the gate structures, wherein after the thinning, the trench isolations are exposed at the back of the semiconductor device, top ends of the vertical-transistor channel structures in each column along the bit-line direction are connected with a line of semiconductor structure extending along the bit-line direction at the backside of the semiconductor memory device to form a bridge-shaped structure, and the trench isolations separate neighboring bridge-shaped structures; recessing, in a first region of the backside of the semiconductor memory device, the trench isolations between the neighboring bridge-shaped structures to expose the gate structures formed in the word-line direction; etching the gate structures formed in the word-line direction, resulting in tunnels along the respective gate structures, the tunnels crossing below each line of semiconductor structure; filling a space where the trench isolations and the gate structures are etched away with a sacrificial material; covering a portion of the first region of the backside of the semiconductor memory device with an insulation layer, the insulation layer enclosing the sacrificial materials below the insulation layer and between the respective neighboring bridge-shaped structures at the covered portion of the first region; and removing the sacrificial material within the space where the trench isolations and the gate structures are etched away while the portion of the first region is covered with the insulation layer, wherein the sacrificial material enclosed by the insulation layer is removed through the tunnels crossing below respective lines of semiconductor structures.

2. The method of claim 1, wherein the insulation layer covers a second region neighboring the portion of the first region covered by the insulation layer, and the trench isolations in the second region are maintained while the trench isolations in the first region are etched.

3. The method of claim 1, wherein the semiconductor memory device has metal shields formed between every two neighboring channel structures along the word-line direction.

4. The method of claim 3, further comprising: etching the metal shields formed in the word-line direction, resulting in tunnels along the respective metal shield crossing below each line of semiconductor structure; filling the space where the trench isolations and the metal shields are etched away with the sacrificial material; and removing the sacrificial material within the space where the trench isolations and the metal shields are etched away while the portion of the first region is covered with the insulation layer, wherein the sacrificial material enclosed by the insulation layer is removed through the tunnels along the respective metal shields crossing below respective lines of semiconductor structures.

5. The method of claim 1, further comprising: covering surfaces of the space where the trench isolations and the gate structures are etched away with a spacer layer before filling the space where the trench isolations and the gate structures are etched away with the sacrificial material.

6. The method of claim 1, further comprises: forming bit-line structures over the lines of semiconductor structures not covered by the insulation layer in the first region while the portion of the first region is covered with the insulation layer.
7. A semiconductor memory device, comprising: trench isolations arranged in a bit-line direction; gate structures arranged in a word-line direction perpendicular to the bit-line direction; an array of vertical-transistor channels arranged in a vertical direction perpendicular to the bit-line direction and the word-line direction and separated by the trench isolations and gate structures, top ends of the array of the vertical-transistor channels in each column being connected to a line of semiconductor structure extending in the bit-line direction at a backside of the semiconductor memory device; and air gap tunnels along the word-line direction that each crosses below the line of semiconductor structure and between two neighboring vertical-transistor channels in a first region at the backside of the semiconductor memory device.
8. The semiconductor memory device of claim 7, wherein the air gap tunnels are adjacent to the gate structures.
9. The semiconductor memory device of claim 7, further comprises metal shields between every two neighboring vertical-transistor channels along the word-line direction.
10. The semiconductor memory device of claim 9, wherein the air gap tunnels are adjacent to the metal shields.
11. The semiconductor memory device of claim 7, wherein a surface of at least one of the air gap tunnels is covered by a spacer layer that surrounds an air gap.
12. The semiconductor memory device of claim 7, wherein top portions of the trench isolation at the backside of the semiconductor memory device are recessed to form air gaps above the trench isolation and between bridge-shaped structures each formed by the respective line of semiconductor structure and the respective vertical-transistor channels in the respective column, and the air gaps are connected with the air gap tunnels.
13. The semiconductor memory device of claim 12, wherein a spacer layer covers surfaces of the air gaps between bridge-shaped structures each formed by the respective line of semiconductor structure and the respective vertical-transistor channels in the respective column and the air gap tunnels connected with the air gap tunnels.
14. The semiconductor memory device of claim 7, wherein an insulation layer covers a portion of the first region and a second region neighboring the portion of the first region, and no air gap tunnel is formed within the second region.
15. A memory system, comprising: a memory controller; and a semiconductor memory device coupled to the memory controller, the semiconductor memory device comprising: trench isolations arranged in a bit-line direction; gate structures arranged in a word-line direction perpendicular to the bit-line direction; an array of vertical-transistor channels arranged in a vertical direction perpendicular to the bit-line direction and the word-line direction and separated by the trench isolations and gate structures, top ends of the array of the vertical-transistor channels in each column being connected to a line of semiconductor structure extending in the bit-line direction at a backside of the semiconductor memory device; and air gap tunnels along the word-line direction that each crosses below the line of semiconductor structure and between two neighboring vertical-transistor channels in a first region at the backside of the semiconductor memory device.
16. The memory system of claim 15, wherein the air gap tunnels are adjacent to the gate structures.
17. The memory system of claim 15, further comprises metal shields between every two neighboring vertical-transistor channels along the word-line direction.
18. The memory system of claim 17, wherein the air gap tunnels are adjacent to the metal shields.
19. The memory system of claim 15, wherein a surface of at least one of the air gap tunnels is covered by a spacer layer that surrounds an air gap.
20. The memory system of claim 15, wherein top portions of the trench isolation at the backside of the semiconductor memory device are recessed to form air gaps above the trench isolation and

between bridge-shaped structures each formed by the respective line of semiconductor structure and the respective vertical-transistor channels in the respective column, and the air gaps are connected with the air gap tunnels, wherein a spacer layer covers surfaces of the air gaps between bridge-shaped structures each formed by the respective line of semiconductor structure and the respective vertical-transistor channels in the respective column and the air gap tunnels connected with the air gap tunnels.